VERIFY: what only the prototype can settle

Every value below was taken from a datasheet, a seller listing or a model. Nobody has measured it yet. Each item says how to check it and what to change if it is off. Work top to bottom: the first two sections gate the rest. Tick an item by noting the measured value and the date here.

1. Before the first power-up (bench, meter, no battery)

#	Item	How	Pass	If not
E1	Battery lead polarity	Meter on the cell's JST-PH plug: red = J2 pin 1 (silk "+")	+ on pin 1	Re-pin the plug (PH crimps lift out with a needle). Never "try it": with USB present a reversed cell is not blocked (round1 m2)
E2	SW1 ON direction (N5)	No battery; bench supply 3.7 V, 100 mA limit, on J2 (+ = pin 1). Slide SW1 both ways and meter TP3 (VSYS) to TP4 (GND)	One direction gives VSYS ≈ 3.7 V	Mark ON on the pack with a dot. If the "ON" side is the awkward one, flip `pack_front` text, not the PCB
E3	Supply sequencing	Bench 3.7 V on J2 (current limit 300 mA), switch ON, MCU flashed, nothing on J3	I < 5 mA; VLED (J3 pin 2) = VSYS once the firmware runs, 0 V while the MCU is held in reset	Check Q2/Q4 orientation and R6/R12
E4	Off current (R17)	Switch OFF, 4.0 V on J2, µA meter in series	≤ 10 µA	Lift Q2 / D3 to find the leak
E5	10 µF 0805 DC bias (N3)	LCR meter at 1 kHz with 4 V bias, or the charger ripple on a scope	≥ 4.7 µF effective	Replace C1/C2 with 1206 or a 25 V 0805
E6	D1 land pattern (N4)	Visual after reflow: fillets on both terminals	Solder fillet both ends	Hand-touch with the iron; widen the pads in v0.2

2. Firmware bring-up (handoff 04)

#	Item	How	Pass	If not
F1	Fuses written (FW-09)	`pio run -e attiny3216 -t fuses`, then flash the debug build and read the boot line	No "BOD OFF" line	Re-run the fuses target; check the programmer wiring
F2	Silicon revision (FW-12)	Debug build prints `SYSCFG.REVID` (add one line), or read it with the programmer	Any revision: the RTC is enabled for the PIT anyway	—
F3	VDD reading (m-9, FW-08)	Debug serial `V=` vs a meter on VSYS at 3.5, 3.8, 4.1 V (bench supply on J2)	Within ±50 mV	Add a calibration factor (measured / read) to `config.h` and scale `vdd_mv()`
F4	Empty sleep and resume	Bench supply on J2 down to 3.25 V for 6 s → fades out and sleeps. Then measure the sleep current. Raise it to 3.8 V or plug in USB	Sleep ≤ 30 µA (LED board unpowered); wakes in 3–5 s	Check PB2/Q4; check the PIT wake (F2)
F5	LED-board recovery (m6)	Unplug and replug J3 with the lamp on	Light returns within ~3 s	Debug log: `tlc_ok` / MODE1

#	Item	How	Pass	If not
F6	Button feel	Press, hold 2 s, auto-repeat	Matches R5–R8	Tune `DEBOUNCE_MS`, `REPEAT_*` in `lightcore.h`

3. Electrical characterisation

#	Item	How	Pass	If not
E7	TLC59116 quiescent current (T0)	All outputs off, VLED 4.20 V: current into J3 pin 2	≤ 20 mA	Recompute the runtimes and the caps with the measured value
E8	Channel current (REXT 750 Ω)	1 Ω in series with one white LED, 100 % PWM, CG 0.992	25.0 mA ± 8 %	Check R1; update `I_CHANNEL_MA` *and* the test
E9	G/B regulation (M1, F10)	Green mode at max, VLED swept 4.2 → 3.4 V. Scope across 1 Ω in series with one G die: read the **peak**, not the average (PWM)	Peak 12.5 mA, held down to VLED ≈ 3.55–3.6 V	Accept, or try REXT 910 Ω (20.6 mA)
E10	Charge current	USB in, cell at ~3.7 V: current into J2	256 mA ± 10 %	Check R3 3.9 k
E11	USB current draw (M4)	USB meter, charging + 4000 K full	Note the value (expect ~0.5–0.6 A)	Use a ≥ 1 A charger (handoff 02)
E12	Unplug handover (m1)	Scope VSYS while pulling USB at 3.6 V battery	Dip ≤ 0.3 V, no reset	Lower R5 further (4.7 k)
E13	Q4 soft turn-on (round2 R2-E1)	Scope VSYS (TP3) and VLED (TP7) at power-on, on the empty-sleep wake, and while hot-plugging J3, at 3.5 V battery	VLED ramps over ~1–2 ms; VSYS dips ≤ 0.2 V; no reset	Raise C8 (220 nF) or add 47 µF bulk on VSYS
E14	F1 PTC when warm (round2 R2-E3)	USB in, charging, 4000 K full for 30 min, pack closed; contact thermometer on F1	No trip; F1 ≤ 60 °C	Move F1 away from U1/D3 in v0.2
E15	VSYS on USB (round2 note)	USB in, 4000 K full: meter TP3; then let the lamp reach empty sleep on a weak USB source	VSYS ≥ 3.9 V; wakes from empty sleep on USB	Lower `VBAT_RESUME_MV` to 3600 in `lightcore.h`

4. Thermal (thermal review §7; thermocouples TC1 board top, TC2 TLC, TC3 PLA skirt, TC4 PLA dome, TC7 cup)

#	Test	Duration	Pass
T1	Each mode at max level, 4.20 and 4.15 V: I_VLED cold and after 30 min	30 min	Within ±8 % of prediction (4000 K full ≈ 182 mA)
T2	Characterisation: derating disabled **on the bench only**, whites at 100 / 150 mA to steady state	~2 h	Gives the real board → PLA offset. Re-fit `T_DERATE_*` if it differs from ~14 K
T3	Release firmware, 4000 K max, on USB, then on battery	2 h each	TC3 ≤ 47 °C, TC4 ≤ 45 °C, TC1 ≤ 70 °C, TC2 ≤ 80 °C, TC7 ≤ 60 °C
T4	T3 in a 30–32 °C box	2 h	TC3 ≤ 50 °C
T5a	`pio run -e bench_t_nack` (sensor NACK) at full output	30 min	Output to 30 % within 3 s; TC3 ≤ 42 °C
T5b	`pio run -e bench_t_stuck` (frozen 25 °C reading)	2 h	Untrusted within 5 min (`T_FROZEN_MS`), current ≤ 110 mA, TC3 ≤ 47 °C

#	Test	Duration	Pass
T6	Hardware thermostat: LEDs at 5 %, hot-air on the board underside to LM75B ≥ 60 °C	10 min	LEDs off within 1 s of OS; resume below 52 °C
T7	Creep: skirt flatness, lug height, skirt OD before and after T3+T4 and 10 × (2 h on / 1 h off)	1 day	Change ≤ 0.05 mm; bayonet still locks firmly

5. Optical and colour

#	Item	How	Pass	If not
O1	PLA transmission (T8a)	Print `coupon_litho_wedge` in the production PLA, backlight it per channel in a dark box	Note τ(t). Contrast visible from 0.9 to 2.6 mm	Change the lithophane range in `moonlib.py` (and re-run the optics model)
O2	Total flux (T8b)	12-direction illuminance at 1 m, 4000 K full, cold	39 lm ± 20 %	If < 31 lm, PLA absorption is higher than assumed: thin the base wall
O3	Page illuminance (T8c, F6)	Lux meter at 25 / 30 cm, 4000 K full, t = 0 and 30 min	20–30 lux expected	Feeds the v0.2 reading-LED decision
O4	Uniformity (T8d)	RAW photos at fixed exposure: WW only, CW only, red only, all white	No visible 4- or 2-fold pattern	Raise the diffuser / board depth (optics model)
O5	Flicker (T9)	Photodiode + scope at levels 0, 3, 8, 15 for each mode; slow fades across CG steps	Only 97.6 kHz content; no visible step at a CG change	Adjust the CG hysteresis in `tlc_output()`
O6	Colour per mode (T10, LED bins)	Spectrometer (or a phone colorimeter app as a rough check)	Whites within 3 SDCM / Duv ±0.003; candle Duv ±0.005	Put the measured xy and lumens into `LED` in `gen_modes.py` and regenerate
O7	Green chromaticity	Same as O6, green only	xy near (0.17, 0.70)	Update `LED["G"]` (520 nm usually sits greener)

6. Mechanical (from the parts' `VERIFY` dicts)

#	Item	Part	Check
M1	Gooseneck end thread M10×1, nut 14 AF × 3.0 mm	`moon_base`, `pack_front`	Calipers on the delivered gooseneck. Change `NUT_AF`, `NUT_T`, `NECK_THREAD_D` in `moonlib.py` / `packlib.py`
M12	Thread protrusion in the moon base (round2 R2-D3)	`moon_base`	With the nut tight, the thread must not stand more than 0.5 mm above the nut top. File it flush; the board's underside is 4.5 mm above
M13	Battery lead path (round2 R2-D18)	`pack_back`, main board slot	Lead exit end and length on the real cell. The plug passes the 6.4 × 5.0 slot; the slack must lie flat in the bay's end gap without crossing the screw heads. If it doesn't, Claude adds a lead channel to `pack_back`
M14	LED-board jig (round2 R2-D12)	`jig_led_board`	The board drops in flush, and the bottom parts don't touch the relief floor
M2	Neck socket 9.3 mm for self-tapping into PETG	`pack_front`	Print `coupon_fit` first
M3	Printed slip fit 0.15 mm radial	all	`coupon_fit`

#	Item	Part	Check
M4	Bayonet: lugs enter, turn 30°, pull down onto a 1.0 mm disc	`moon_shell`, `moon_base`	`coupon_bayonet`
M5	Ring preload: board must not rattle with the moon locked	`moon_ring`	Add a 0.2 mm shim ring if it rattles
M6	LED board thickness 1.0 mm as ordered	`moon_base`	Calipers on the delivered board
M7	Battery 64.0 × 35.5 × 5.3 mm (listing)	`pack_back`	Calipers on the delivered cell; bay is 36.0 × 64.5 × 5.6 + 0.6 swell
M8	USB-C receptacle centre 1.63 mm above the PCB; PCM12 knob 0.9 mm above, 1.0 mm past the edge; J2 header 5.5 mm tall (JST), under a 0.8 mm face pocket; your USB-C cables seat fully in the 13.4 × 7.6 window	`pack_front`	Calipers on the assembled main board before printing the shell; try your cables in the printed shell
M9	Clip force ~5 N at 3 mm	`pack_clip`	Holds on a hardback and a paperback without marking them
M10	Button caps 7.0 mm in 7.4 mm holes	`button_caps`	Slide freely; sand if tight
M11	Lithophane 0.9–2.6 mm reads well at 0.12 mm layers	`moon_shell`	First print, lit and unlit